

CENTRAL BOARD OF SECONDARY EDUCATION — PRACTICE EXAMINATION
CLASS IX • SCIENCE (SUBJECT CODE: 086) • ACADEMIC SESSION 2026-27

SET E — CASE STUDY & PRACTICAL EXPERIMENTS

CHAPTER 08: JOURNEY INSIDE THE ATOM: STRUCTURE & COMPOSITION
TIME ALLOWED: 45 MINUTES MAXIMUM MARKS: 25 MARKS SERIES: CBS-IX-SCI-SET-E-08

Student's Name: _____

Roll No.: _____

Class & Section: Class IX - _____

Date of Exam: _____

Invigilator's Sign: _____

GENERAL INSTRUCTIONS (SET E - CASE STUDY & PRACTICAL EXPERIMENTS):

1. The question paper comprises **5 In-Depth Practical Case Studies** with **5 MCQs each (Total: 25 Questions, 1 Mark each)**.
2. Read each laboratory experiment, data table, and passage carefully before answering the corresponding questions.
3. Each question has 4 options with a single correct answer.
4. All questions are compulsory. There is no negative marking.
5. Darken the corresponding bubble on the **OMR Sheet** completely using a blue/black ballpoint pen.

CASE STUDY 1: GEIGER-MARSDEN ALPHA PARTICLE SCATTERING EXPERIMENT

In 1911, under the direction of Ernest Rutherford, Hans Geiger and Ernest Marsden conducted their landmark alpha particle scattering experiment.

- A radioactive bismuth source emitted a narrow beam of high-energy positively charged alpha (α) particles ($m \approx 4 \text{ u}$, charge $+2e$).
- The α -beam struck an extremely thin gold foil of thickness approximately 100 nm (10^{-7} m), about 1000 atoms thick).
- A movable zinc sulfide (ZnS) phosphorescent detector screen and microscope recorded scintillations produced by scattered particles at various angles (θ).
- **Observations:**
 1. Over 99.8% of the α -particles passed straight through the foil with zero or negligible deflection.
 2. A small fraction (about 0.14%) were deflected by angles greater than 1° .
 3. Only about 1 in 12,000 particles suffered massive deflection ($> 90^\circ$) or rebounded backwards at nearly 180° .

Q.1 What major atomic structural conclusion did Rutherford deduce from the observation that most α -particles passed undeflected?

- | | |
|---|---|
| (A) Atoms are solid indivisible spheres like billiard balls | (B) Most of the internal space within an atom is completely empty |
| (C) Electrons are located at the center of the atom | (D) Gold atoms contain zero electrons |

Q.2 Why did Rutherford deduce that the positive charge and nearly all atomic mass reside in a tiny concentrated 'nucleus'?

- | | |
|---|---|
| (A) Because alpha particles were attracted towards the center | (B) Because only a massive, highly concentrated positive core could exert sufficient electrostatic Coulomb repulsion to reverse the trajectory of fast alpha particles at 180° |
| (C) Because electrons are positively charged | (D) Because gold is yellow in color |

Q.3 Why was gold chosen as the target foil metal rather than aluminum or iron?

- (A) Gold is radioactive (B) Gold is the most malleable of all metals, allowing hammering into ultra-thin sheets only a few hundred atoms thick
- (C) Gold has no electrons (D) Gold does not conduct electricity

Q.4 Rutherford estimated that the radius of the nucleus is approximately how many orders of magnitude smaller than the radius of the atom?

- (A) 10 times (B) 100 times
- (C) 10^5 times (100,000 times) (D) 10^{10} times

Q.5 What was the fatal theoretical limitation of Rutherford's planetary model of the atom according to classical physics?

- (A) It could not explain radioactivity (B) According to classical electrodynamics, an accelerating orbiting electron should continuously radiate energy and spiral into the nucleus in 10^{-8} s, making atoms unstable
- (C) It predicted that electrons have positive charge (D) It violated the law of conservation of mass

CASE STUDY 2: CATHODE RAYS, CANAL RAYS & SUBATOMIC PARTICLE DISCOVERY

In the late 19th and early 20th centuries, physicists explored electrical discharge through rarefied gases in evacuated glass discharge tubes (Crookes tubes):

- **J.J. Thomson's Cathode Ray Experiment (1897):** High voltage (10 kV) applied across electrodes at low pressure (0.001 mm Hg) generated rays travelling from cathode to anode. The rays cast sharp shadows of obstacles, rotated light mica paddle wheels, and deflected towards the positive plate in an electric field. The charge-to-mass ratio was invariant ($e/m = 1.76 \times 10^{11} \text{ C/kg}$).
- **E. Goldstein's Canal Ray Experiment (1886):** Using a perforated cathode, luminous rays travelled in the opposite direction from anode to cathode, carrying positive charges whose e/m ratio depended on the gas in the tube.

Q.6 Which subatomic particle was discovered by J.J. Thomson through cathode ray investigations?

- (A) Proton (B) Neutron
- (C) Electron (D) Positron

Q.7 What experimental evidence proved that cathode rays possess mechanical kinetic energy and momentum?

- (A) They cause fluorescence on ZnS screens (B) They can mechanically rotate a delicate mica paddle wheel placed in their path
- (C) They cast sharp shadows (D) They penetrate thin paper

Q.8 Why was the charge-to-mass ratio (e/m) of canal rays dependent on the specific gas present in the discharge tube?

- (A) Because canal rays are not fundamental particles (B) Because canal rays are residual gas cations formed when gas atoms lose electrons, and different gas ions possess different atomic masses
- (C) Because electrons change mass in different gases (D) Because the voltage fluctuated

Q.9 Which gas in the discharge tube yielded positive ions with the highest e/m ratio (lightest mass), leading to the identification of the proton?

- (A) Helium (B) Hydrogen gas (H_2)
- (C) Neon (D) Oxygen

Q.10 Who discovered the third fundamental subatomic particle, the neutron, in 1932 by bombarding beryllium with alpha particles?

- | | |
|--------------------|---------------------|
| (A) James Chadwick | (B) Niels Bohr |
| (C) John Dalton | (D) Robert Millikan |

CASE STUDY 3: BOHR-BURY ELECTRONIC CONFIGURATIONS & ENERGY SHELL QUANTUM LEVELS

To resolve the instability of Rutherford's atomic model, Niels Bohr proposed quantized stationary orbits in 1913. The Bohr-Bury scheme governs the arrangement of electrons in shells (designated K, L, M, N... with principal quantum numbers $n = 1, 2, 3, 4$):

1. Maximum electrons in shell $n = 2n^2$.
 2. Maximum electrons in the outermost valence shell = 8 (Octet rule).
 3. Electrons do not occupy a new shell until the inner shells are completely filled (stepwise filling).
- Sodium ($Z=11$): Electronic configuration $2, 8, 1$
 - Argon ($Z=18$): Electronic configuration $2, 8, 8$
 - Potassium ($Z=19$): Electronic configuration $2, 8, 8, 1$
 - Calcium ($Z=20$): Electronic configuration $2, 8, 8, 2$

Q.11 What is the maximum theoretical electron capacity of the M-shell ($n=3$) according to the $2n^2$ rule?

- | | |
|--------|--------|
| (A) 8 | (B) 18 |
| (C) 32 | (D) 2 |

Q.12 Why does potassium ($Z=19$) have the configuration $2, 8, 8, 1$ rather than $2, 8, 9$?

- | | |
|--|---|
| (A) Because M-shell capacity is only 8 | (B) Because according to the Bohr-Bury rule, the outermost valence shell cannot hold more than 8 electrons; hence, the 19th electron enters the N-shell |
| (C) Because potassium is an inert gas | (D) Because potassium has no neutrons |

Q.13 What happens when an electron absorbs a discrete quantum of energy (photon)?

- | | |
|----------------------------------|---|
| (A) It spirals into the nucleus | (B) It jumps (is excited) from a lower stationary energy level to a higher permissible energy level |
| (C) It splits into two positrons | (D) Its mass doubles |

Q.14 Why do noble gases like Argon ($2, 8, 8$) exhibit chemical inertness under standard conditions?

- | | |
|--------------------------------------|--|
| (A) They have zero protons | (B) Their valence shell contains a complete stable octet of 8 electrons, requiring extremely high energy to gain or lose electrons |
| (C) They exist only at absolute zero | (D) They lack mass |

Q.15 Which subatomic particle distribution represents the neutral atom of Aluminum ($Z=13$, Mass number $A=27$)?

- | | |
|--|--|
| (A) 13 protons, 14 neutrons, 13 electrons (config: $2, 8, 3$) | (B) 14 protons, 13 neutrons, 14 electrons (config: $2, 8, 4$) |
| (C) 13 protons, 27 neutrons, 13 electrons (config: $2, 8, 3$) | (D) 27 protons, 13 neutrons, 27 electrons |

CASE STUDY 4: VALENCY DYNAMICS & CHEMICAL FORMULA FORMULATION

Students tabulated atomic numbers, electron configurations, valence electrons, and valency for elements from $Z=1$ to $Z=17$ to understand chemical bonding:

- **Element P (Lithium, $Z=3$):** Config $2, 1 \rightarrow$ Valence electrons = 1 $\rightarrow$ Valency = 1
- **Element Q (Nitrogen, $Z=7$):** Config $2, 5 \rightarrow$ Valence electrons = 5 $\rightarrow$ Valency = $8 - 5 = 3$
- **Element R (Magnesium, $Z=12$):** Config $2, 8, 2 \rightarrow$ Valence electrons = 2 $\rightarrow$ Valency = 2
- **Element S (Chlorine, $Z=17$):** Config $2, 8, 7 \rightarrow$ Valence electrons = 7 $\rightarrow$ Valency = $8 - 7 = 1$

Q.16 How is the valency of non-metallic elements with 4 or more valence electrons (such as Nitrogen and Chlorine) determined?

- (A) Valency = Number of valence electrons
 (B) Valency = $8 - (\text{Number of valence electrons})$
 (C) Valency = Atomic number divided by 2
 (D) Valency = Mass number minus 8

Q.17 What is the chemical formula of the compound formed between Element R (Magnesium, valency 2) and Element S (Chlorine, valency 1)?

- (A) RS
 (B) RS_2 ($MgCl_2$)
 (C) R_2S (Mg_2Cl)
 (D) R_2S_3

Q.18 What is the chemical formula of the compound formed between Element R (Magnesium, valency 2) and Element Q (Nitrogen, valency 3)?

- (A) R_3Q_2 (Mg_3N_2 , Magnesium nitride)
 (B) RQ (MgN)
 (C) R_2Q_3
 (D) RQ_2

Q.19 Why does Helium ($Z=2$, config 2) have a valency of 0, whereas Magnesium ($Z=12$, config $2, 8, 2$) has a valency of 2?

- (A) Helium has an unstable nucleus
 (B) For Helium ($n=1$), its single K-shell has maximum capacity of 2 (stable duplet), whereas Magnesium has 2 easily lost valence electrons outside a stable octet
 (C) Helium is a solid metal
 (D) Magnesium has zero neutrons

Q.20 What type of chemical bond is formed when Magnesium transfers its 2 valence electrons to Oxygen ($Z=8$, config $2, 6$)?

- (A) Covalent bond
 (B) Ionic (electrovalent) bond
 (C) Metallic bond
 (D) Hydrogen bond

CASE STUDY 5: ISOTOPES, ISOBARS & APPLICATIONS IN MEDICINE AND INDUSTRY

Mass spectrometry reveals that many chemical elements occur in nature as mixtures of stable isotopes:

- **Chlorine:** Consists of $^{35}_{17}Cl$ (atomic mass 35 u , abundance 75%) and $^{37}_{17}Cl$ (atomic mass 37 u , abundance 25%).
- **Hydrogen:** Occurs as Protium (1_1H), Deuterium (2_1H), and Tritium (3_1H).
- **Isobars:** Argon ($^{40}_{18}Ar$) and Calcium ($^{40}_{20}Ca$) possess identical mass numbers ($A=40$) despite having different atomic numbers ($Z=18$ and $Z=20$).

Radioactive isotopes (radioisotopes) find extensive use in medicine, agriculture, and nuclear power.

Q.21 Calculate the average relative atomic mass of natural chlorine:

- (A) 35.0 u
 (B) 35.5 u
 (C) 36.0 u
 (D) 37.0 u

Q.22 How do isotopes of an element differ in their subatomic composition?

- (A) They have different numbers of protons
(B) They have identical numbers of protons and electrons, but different numbers of neutrons
(C) They have different electron configurations
(D) They have different nuclear charges

Q.23 Why do all isotopes of chlorine exhibit identical chemical reactivity with sodium?

- (A) Because they have the same mass number
(B) Because chemical reactivity is governed exclusively by valence electrons and electronic configuration, which are identical across isotopes
(C) Because neutrons react with sodium
(D) Because both isotopes are radioactive

Q.24 Which radioisotope is universally utilized in medical radiotherapy to treat malignant cancerous tumors?

- (A) Cobalt-60 (^{60}Co)
(B) Carbon-14 (^{14}C)
(C) Uranium-238 (^{238}U)
(D) Sodium-23 (^{23}Na)

Q.25 Which radioisotope is used in clinical medicine for the diagnosis and treatment of goitre and thyroid gland disorders?

- (A) Iodine-131 (^{131}I)
(B) Cobalt-60 (^{60}Co)
(C) Uranium-235 (^{235}U)
(D) Phosphorus-32 (^{32}P)

CBSE CANDIDATE OMR ANSWER RESPONSE GRID (SET E)

01 (A) (B) (C) (D)	02 (A) (B) (C) (D)	03 (A) (B) (C) (D)	04 (A) (B) (C) (D)	05 (A) (B) (C) (D)
06 (A) (B) (C) (D)	07 (A) (B) (C) (D)	08 (A) (B) (C) (D)	09 (A) (B) (C) (D)	10 (A) (B) (C) (D)
11 (A) (B) (C) (D)	12 (A) (B) (C) (D)	13 (A) (B) (C) (D)	14 (A) (B) (C) (D)	15 (A) (B) (C) (D)
16 (A) (B) (C) (D)	17 (A) (B) (C) (D)	18 (A) (B) (C) (D)	19 (A) (B) (C) (D)	20 (A) (B) (C) (D)
21 (A) (B) (C) (D)	22 (A) (B) (C) (D)	23 (A) (B) (C) (D)	24 (A) (B) (C) (D)	25 (A) (B) (C) (D)

Total Questions Attempted: ____ / 25

Marks Obtained: ____ / 25

Evaluator's Signature: _____

OFFICIAL ANSWER KEY & PRACTICAL INVESTIGATION EXPLANATIONS (SET E)

Q. No.	Correct	Scientific Rationale & Experimental Data Analysis
Q.1	(B)	Since over 99.8% of alpha particles traversed the foil undeflected, Rutherford concluded that the vast majority of atomic volume is empty space.
Q.2	(B)	To deflect a fast, massive alpha particle backwards, the positive charge and mass must be concentrated in a minuscule, dense central nucleus.
Q.3	(B)	Malleability allows fabrication of foils ~100 nm thick, ensuring alpha particles encounter at most a single atomic collision.
Q.4	(C)	Atomic radius is $\sim 10^{-10}$ m; nuclear radius is $\sim 10^{-15}$ m. The ratio is $10^{-10} / 10^{-15} = 10^5$.
Q.5	(B)	Accelerating charges radiate electromagnetic waves. An orbiting electron would lose kinetic energy and collapse into the nucleus, contradicting atomic stability.
Q.6	(C)	Thomson proved cathode rays are streams of negatively charged fundamental particles, which were named electrons.
Q.7	(B)	Rotating a paddle wheel demonstrates that cathode rays consist of particles possessing mass and momentum ($p = mv$).
Q.8	(B)	Canal rays are positive ions (M^{+}). Since hydrogen, helium, and nitrogen ions have different atomic masses, their e/m ratios vary.
Q.9	(B)	Hydrogen produces the lightest positive ion (H^{+}), having charge $+e$ and mass $\approx 1 \text{ u}$, identified as the proton.
Q.10	(A)	James Chadwick discovered the uncharged neutron (1_0n) via the reaction ${}^9_4\text{Be} + {}^4_2\alpha \rightarrow {}^{12}_6\text{C} + {}^1_0n$.
Q.11	(B)	Capacity $= 2n^2 = 2(3)^2 = 2 \times 9 = 18$ electrons.
Q.12	(B)	Rule 2 dictates that the outermost shell cannot exceed 8 electrons. Once the M-shell reaches 8, additional electrons begin filling the N-shell.
Q.13	(B)	Electrons absorb energy to transition to higher discrete energy states (excitation) and emit photons when returning to lower ground states.
Q.14	(B)	A complete valence octet confers maximal thermodynamic stability, preventing spontaneous chemical reactions.
Q.15	(A)	Atomic number $Z = 13$ (13 protons, 13 electrons with config $2, 8, 3$). Neutrons $N = A - Z = 27 - 13 = 14$.
Q.16	(B)	Non-metals complete their octet by gaining or sharing $(8 - V)$ electrons, where V is the number of valence electrons.
Q.17	(B)	Criss-crossing valencies (Mg^{2+} and Cl^{-1}) yields $MgCl_2$.
Q.18	(A)	Criss-crossing valencies (Mg^{2+} and N^{3-}) gives Mg_3N_2 (Magnesium nitride).
Q.19	(B)	K-shell capacity is 2; Helium has a completely filled shell (duplet), giving valency 0. Magnesium has 2 valence electrons in M-shell, which it loses to form Mg^{2+} .
Q.20	(B)	Complete electron transfer from a metal ($Mg \rightarrow Mg^{2+} + 2e^{-}$) to a non-metal ($O + 2e^{-} \rightarrow O^{2-}$) forms an ionic electrovalent bond.
Q.21	(B)	$\text{Average Mass} = \frac{(35 \times 75) + (37 \times 25)}{100} = \frac{2625 + 925}{100} = \frac{3550}{100} = 35.5 \text{ u}$.
Q.22	(B)	Isotopes share the same atomic number Z (same protons and electrons) but differ in neutron count ($N = A - Z$).

Q. No.	Correct	Scientific Rationale & Experimental Data Analysis
Q.23	(B)	Chemical bonding involves valence electrons. Since all isotopes have identical electronic structure (2, 8, 7), their chemical properties are identical.
Q.24	(A)	Cobalt-60 emits high-energy penetrating gamma rays (γ) that destroy rapidly dividing malignant cancer cells.
Q.25	(A)	The thyroid gland selectively absorbs iodine from the bloodstream. Iodine-131 is used as a radiotracer and therapeutic agent for thyroid disorders.